

WHITEPAPER

UNDERSTANDING THE CRYPTOASSET MARKET



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The technological advancement behind cryptoassets have come a long way and have the potential to disrupt the financial system as we know it.

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Executive Summary

Cryptoassets represent a seismic shift in the financial markets and, in recent years, have grown in popularity. The technological advancement behind cryptoassets have come a long way and have the potential to disrupt the financial system as we know it.

Central banks and other financial institutions can play a key role in shaping this landscape. However, this global phenomenon is creating confusion on multiple levels i.e. how individual cryptoassets differ from one another and the role of the major participants in the cryptoasset ecosystem. With the buzz around bitcoin, altcoins, cryptocurrencies and tokens, a whole new financial ecosystem has been created. In this whitepaper, we will discuss the taxonomy of cryptoassets, market participants and the current capital market landscape.



Introducing Cryptoassets

Cryptoassets are digital assets which use cryptographic techniques to generate a medium of exchange of financial transactions. Cryptocurrencies, utility coins, security tokens are all different types of cryptoassets. A cryptocurrency is a digital or virtual currency. The currency is encrypted (secured) using cryptography to secure financial transactions, create additional units, and verify the transfer of assets. In contrast to fiat currency and central banking systems, many cryptocurrencies are decentralised systems based on blockchain technology.

Taxonomy of Cryptoassets

Cryptoassets are a superset of cryptocurrencies. The most widely known cryptocurrency is Bitcoin. Altcoins and crypto tokens are subset categories of cryptocurrency. Litecoin, Bitcoin Cash and Dogecoin are common examples of altcoins. Crypto tokens are special cryptoasset tokens that reside on their own blockchains and represent an asset or utility [1].

Bitcoin and altcoins are virtual currencies that have dedicated blockchains and are primarily used as a medium for digital payments. Crypto tokens, however, operate on top of a blockchain that acts as a medium for creation and execution of decentralised apps and smart contracts and are used to facilitate transactions. In the following subsections, we briefly introduce virtual currency products (including Bitcoin and Ether), utility tokens, security tokens and stablecoins.

Convertible Virtual Currency Products

The U.S. Department of the Treasury, through Financial Crimes Enforcement Network (FinCEN), defines 'real' currency as a jurisdiction's coin and paper money that is designated as legal tender and that is customarily used and accepted as a local medium of exchange. A 'virtual' currency is "a medium of exchange that operates like a currency in some environments, but does not have all the attributes of real currency." 'Convertible' virtual currency is a virtual currency that has "an equivalent value in real currency, or act[s] as a substitute for real currency." Virtual currencies and convertible virtual currencies are not legal tender in any jurisdiction [2].

Bitcoin was created as a reward for payment processing using cryptography, known as mining, in which participants offer their computing power to verify and record payments into a public ledger. Read more on Blockchain technologies [3].

With a goal of building decentralised applications, Vitalik Buterin proposed the development of a new platform with a more general scripting language for application development. This led to the origination of Ethereum, a decentralised platform that runs smart contracts, and Ether, a form of payment made by clients of the platform.

Utility Tokens

Utility tokens are a digital token of cryptocurrency issued to fund the development of an ecosystem in which the utility token, at a later date, can be used to purchase goods or services offered by the issuer. For example, by creating utility tokens, a startup can sell 'digital coupons' for the service it is developing. Utility token creators usually refer to these crowd sales as token generation events (TGEs) or token distribution events (TDEs) to avoid the appearance that they are engaging in a securities offering [4]. Utility tokens do not primarily function as a speculative instrument with fluctuating market price.



Digital Security

Digital Security is considered an investment contract. They are subject to securities regulations and offer an array of possible applications as financial instruments. The main use-case for these type of tokens is the ability to offer a digital representation of shares of a company's stock in anticipation of future profits e.g. in the form of dividends. These differ from utility tokens which are offered by a start-up to its client for the purpose of financing the client's future purchases.

Stablecoins

Stablecoin is a subset of cryptoassets backed by a real-world asset(s) e.g. dollar, gold, bitcoin, hybrid assets, etc. Stablecoins are price-stabilised and mirror the volatility profiles of reference assets. They incorporate Bitcoin or Ether features: programmability (e.g. smart contract integration), efficiency (e.g. low-to-zero transaction fees, fast settlement times), fungibility and open or permissionless access [5]-[6]. Stablecoins also provide a liquidity solution for cryptoasset exchanges as most exchanges can't accept dollar, euro or other fiat deposits.

There are two main types of stablecoins: algorithmic and asset-backed. Of the 57 stablecoins reviewed in the "Global Cryptoasset Benchmarking Study," [7]

algorithmic represents 23%, and asset-backed represents 77%, of which 33% is made up of traditional collateral and 44% crypto collateral. The algorithmic stablecoins employ a set of rules, expressed in software code, that attempt to match the supply of the stablecoin with demand. In comparison, an asset-backed stablecoin design is one where some asset is held in reserve to support the stablecoin's exchange rate.

The US dollar is the most common stability benchmark and is utilised by 66% of all stablecoins. Other benchmarks include other fiat currencies (e.g. euro, yen), commodities (e.g. gold), and inflation (e.g. G10 average country inflation). Exchange of cryptocurrency for sovereign currency through a third-party exchanger is generally money transmission and requires Money Transmitters Licenses (MTLs) and the issuers and administrators potentially need to register as Money Service Businesses (MSB) as mandated by FinCen.

Stablecoin issuers need to be aware of securities law to avoid potential regulatory risk. In December 2018, the stablecoin Basis shut down its operations and returned all of its remaining funds to investors. Basis uses bond and share tokens to expand and contract the supply of tokens in order to maintain a price peg. Therefore, it is limited to accredited investors and requires know-your-customer (KYC) and anti-money laundering (AML) checks.

Major Cryptoasset Participants

A cryptoasset ecosystem, composed of a diverse set of actors, builds interfaces between public blockchains, traditional finance and various economic sectors. These services add significant value to cryptoassets as they provide the means for public blockchains and their native currencies to be used in the broader economy [7]. There are four major cryptoasset participants today, namely exchanges, banks, miners and custodians.

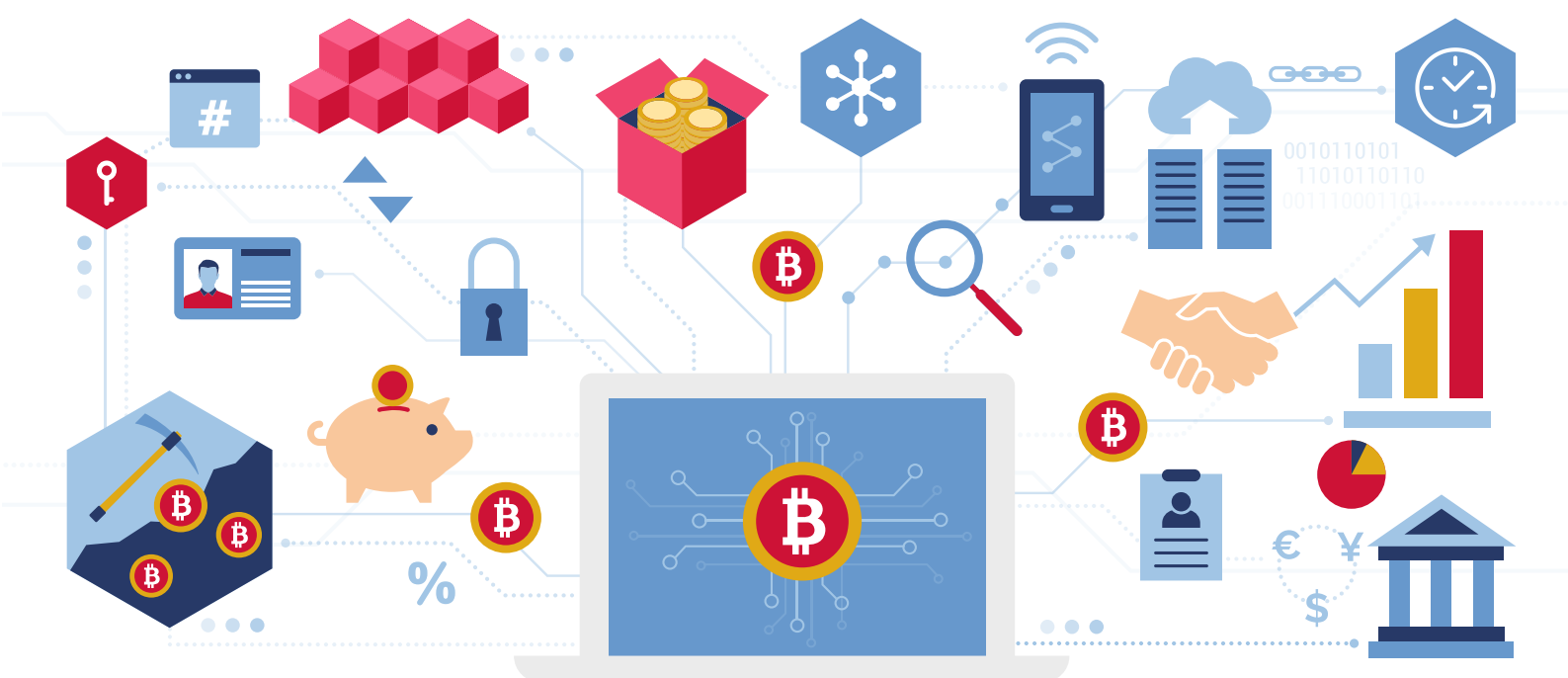
Exchanges

A cryptoasset exchange allows participants to trade cryptoassets. It provides an online matching platform for supply and demand. There are currently over 210 exchanges listed, most of which are centralised, meaning there is a trusted middleman to handle the trades. Based on volume and estimated revenues, the top three cryptoasset exchanges are centralised cryptoasset exchanges (CEXs), such as Binance, Huobi and OKEX.

Banks serve as a key gateway for converting fiat to cryptoasset in the digital ecosystem. Table 1 shows a representative list of cryptoasset marketplaces and their associated banking partners. The dynamic bank partnership structure in the current ecosystem reflects the fast-evolving regulatory development for banks and as well as CEXs.

Table 1: Exchanges and Their Associated Banks

Exchange	Bank
Bitfinex	Deltec Bank, Bank of Communications
bitFlyer	Silvergate Bank
Bithump	Nonghyup Bank
Bitstamp	Gorenjska Banka, Unicredit
Bittrex	Signature Bank
Coinbase	Metropolitan Commercial Bank, Silvergate Bank, LHV Bank, Barclays
Coinone	Nonghyup Bank
Gemini	Silvergate Bank
Kraken	Silvergate Bank, Fidor Bank, Evolve Bank
Robinhood Crypto	Apex (BMO Harris Bank)
Upbit	Shinhan Bank



A decentralised cryptoasset exchange (DEX) like Etherdelta, IDEX and HADAX cut out the middleman by creating a highly intelligent trustless environment which facilitates peer-to-peer cryptoasset trading that does not store users' funds on the exchange.

Miners

Like mining for gold, Cryptoassets are mined by individuals and companies using computers to process transactions and earn a cryptoasset reward. This is a very resource-intensive process, requiring heavy computing power to satisfy security conditions (rooted in cryptography) and ensure that all network participants agree the blockchain is accurate. Mining has spawned many sub-industries including mining pools such as BTC.com, Antpool, ViaBTC [8], hardware equipment manufacturers such as Bitmain, Avalon, Nvidia, and AMD GPU and mining cloud services such as Genesis Mining, Hashnest, Hashflare [9].

Banks

The opportunities for banks to utilise blockchain technologies could have far-reaching implications. However, direct cryptoasset trading is relatively limited for banks. Areas in which banks could potentially leverage blockchain technologies include payments, clearing, settlement, KYC and AML processes and trade finance amongst others.

Custodians

Cryptoasset custody solutions provide storage and security services. They incorporate a combination of hot storage (with a connection to the Internet), cold storage (disconnected from the Internet) and vault storage (a combination of both). Coinbase and Fidelity Investments are two firms currently offering or designing these custody services.

Cryptoasset Capital Market

Recently there has been substantial growth in cryptoasset marketplaces and product offerings. The marketplace is complex and comprises of the traditional stock market, foreign exchange and the cryptoasset market. As outlined in Diagram 1, this section of the paper provides details on both crypto-only and fiat-supporting markets. We also explore trading venues, such as centralised and decentralised exchanges and the over-the-counter (OTC) markets.

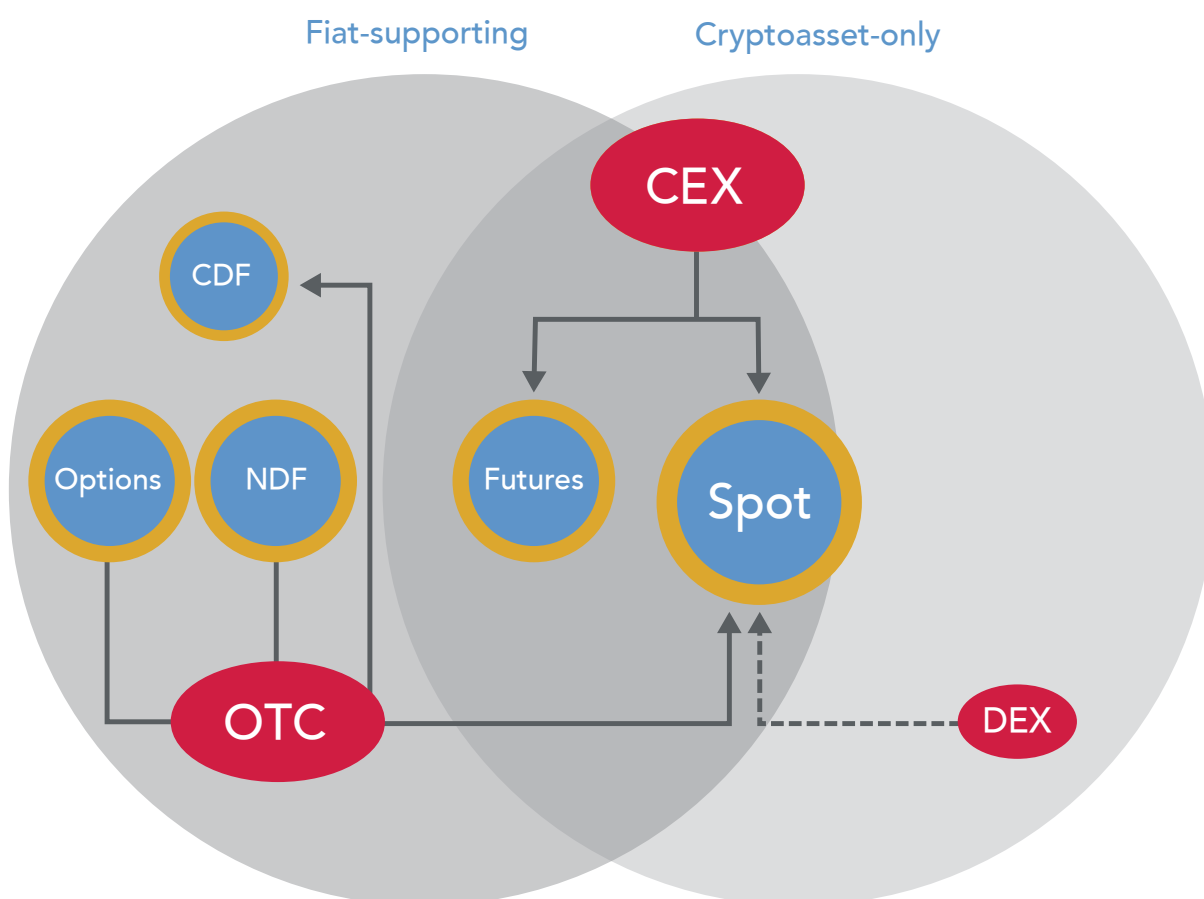


Diagram 1: Overview of Cryptoasset Capital Market

Cryptoasset-Only Market

Cryptoasset exchanges are central to the development of the capital market. Currently, there are over 210 exchanges with cryptoasset trading offers. However, over 95% of the total daily trading volume is processed by the top 50 exchanges, all of which are centralised exchanges (CEXs). Cryptoasset-only marketplaces allow users to trade one cryptoasset for another to gain diversification within the cryptoasset complex.

Centralised Exchanges (CEXs)

Centralised exchanges (CEXs) were formed during the emergence of the cryptoasset industry in 2010. Most cryptoasset exchanges are centralised and privately owned, whilst some are registered in a specific jurisdiction, such as Binance and Huobi.

A cryptoasset exchange is mainly composed of order placement, order cancellation and order matching functions. For CEXs, the exchange architecture relies upon third-party service providers. This however, can lead to single point failure. For instance, the purchase and sale of cryptoasset between members of an exchange does not involve the transfer of ownership of the asset too. Therefore, users of the CEXs are actually operating within virtual accounts on the exchange, which have ultimate ownership of the cryptoasset. In other words, the trades on CEXs are done on an IOU basis e.g. a debt instrument. The users of the CEXs do not receive their funds until they explicitly withdraw their funds from the exchange. Many CEXs were not prepared for the influx of users they received, and as a result, they are vulnerable to major system failures and cybersecurity risk.

Decentralised Exchanges (DEXs)

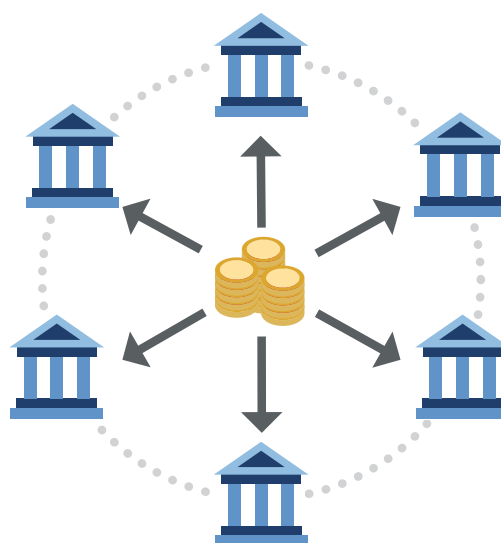
On the other hand, decentralised exchanges (DEXs) aim to broadcast each part of the transactions to a decentralised network such that no centralised third-party server is needed in the trading system. DEXs' architecture has no central controlling server and does not rely on third-party services to store user funds. Through an automated process on blockchain, peer-to-peer transactions are stored in their personal wallets. DEXs provide greater confidentiality and reduce the risk of server inaccessibility and single point of failure.

A number of DEXs are hybrids of CEXs and DEXs, which allow for fast access to centralised gateways. This type of architectural design includes on-chain operations that do not require significant network capacities, such as settlement. However, the order books can be established through off-chain trade

entry, using the centralised gateways. Another type of DEXs is based on Ethereum smart contracts or the mechanics of multi-sig signatures, providing high security for cryptoasset transactions. However, such DEXs suffer from slow transaction speed and are not efficient cryptoasset marketplaces.

In summary, DEXs facilitate value flow and align with the decentralised values espoused by the blockchain community. However, DEXs suffer from slow transaction speed and are not yet efficient digital asset marketplaces.

Both CEXs and DEXs need to operate under the rules and regulation set forth by regulators and law enforcement agencies. On November 8th, 2018, the Securities and Exchange Commission (SEC) settled charges against Zachary Coburn, the founder of EtherDelta, a digital 'token' trading platform. According to the SEC's order, EtherDelta is an online platform for secondary market trading of ERC20 tokens, a type of blockchain-based token commonly issued in Initial Coin Offerings (ICOs). This is the SEC's first enforcement action based on findings that such a platform operated as an unregistered national securities exchange. Here we note that none of the cryptoasset exchanges fit the SEC definition of 'national securities exchange.' They are merely marketplaces.



Fiat-supporting Market

Fiat-supporting marketplaces are governed by rules and regulations based on jurisdiction and the type of services each exchange offers, which result in a different market structure and ecosystem. The majority of trading volumes on fiat-supporting marketplaces represents continuous on-off ramp fiat money flow from the cryptoasset ecosystem to the incumbent financial system and the real economy. In the United States, exchanges, such as Coinbase and Robinhood, are regulated through FinCen's Money Transmission Laws to combat money laundering activities for 'convertible virtual currencies'. All cryptoasset exchanges suffer from regulatory uncertainty and this is especially true for fiat-supporting marketplaces. As the regulatory framework solidifies and provides investors with protection, such marketplaces will grow.

Exchange-Traded Products

Spot: Exchange-traded spot products are the largest segment of the 'on-off ramp' or 'fiat-to-crypto' segment by trading volume, which is much smaller than the cryptoasset-only spot trading volume. Currently, no 'fiat-supporting' DEXs exist due to KYC and AML considerations [10].

Futures: Futures contracts enable investors to take short positions. Bitcoin futures contracts started trading on CME and the Chicago Board Options Exchange on 10th December 2017. These products allow investors to hedge or speculate on bitcoin prices without actually transacting in the cryptocurrency.

Futures contracts rely on a robust and representative underlying instrument that the contract will be priced against. However, there is no centralised price or exchange for Bitcoin. The CME and CBOE chose different methods for calculating the underlying price for their futures contracts. The CME bitcoin futures contract employs a Bitcoin Reference Rate (BRR), which is calculated and published by Crypto Facilities Ltd.

(CF). The CBOE chose to reference an end-of-day auction rate published by Gemini, a prominent cryptocurrency marketplace.

CME set the required margin level at 35% of notional value for Bitcoin contracts, whilst CBOE set the required margin level at 40%. CME Bitcoin futures contract size is 5 Bitcoins per contract whereas it is 1 Bitcoin for CBOE Bitcoin futures contract. The larger contract size for Bitcoin futures instruments allows institutional traders to trade and risk manage positions efficiently in OTC markets.

Over-the-Counter (OTC) Market

Spot: The cryptoasset market is fragmented and competing exchanges exacerbate the liquidity profile of products traded on an exchange. For institutional traders seeking anonymity and liquidity for large block trades, the bilateral OTC market has emerged as a feasible trading venue to minimise slippage and market impact.

Given the significant counterparty and credit risk associated with the bilateral nature of OTC transactions, the OTC marketplace is slowly embracing regulation as a way to build trust and create a healthy and fair environment. For instance, Genesis Trading is a broker-dealer registered under Securities Exchange Act of 1934 and is a member of the Financial Industry Regulatory Authority (FINRA). It is also subject to ongoing supervision by New York State Department of Financial Services (NYDFS) under a BitLicense. Other liquidity providers could also be Swap Exchange Facilities (SEFs), which provide an organised framework for dealing in eligible OTC instruments with obligatory post-trade reporting.

As the cryptocurrency market matures, we expect OTC trades to dominate volumes on exchanges due to the emergence of diverse derivatives products for institutional trading and hedging.

Non-Deliverable Forward (NDF): Non-deliverable forward (NDF) instrument has existed in emerging market foreign exchange rate with capital controls for decades. Taking the design of emerging market foreign exchange NDFs, TeraExchange was the first regulated marketplace to offer U.S. Dollar settled Bitcoin forwards. Bitcoin, in this case, is analogous to the emerging market foreign exchange rate.

The notional amount of the Bitcoin NDF is never exchanged. The only exchange of cash is the difference between the Bitcoin forward price at trade date and the prevailing Tera Bitcoin Price Index ('Index Price') on the valuation date.

Contract for Difference (CFD): CFD products are prohibited for U.S. residents and citizens, but permissible in other jurisdictions, such as the European Union. Through CFDs, traders can participate in the price movements of Bitcoin without actually owning the underlying asset; in this case, Bitcoin. When trading the CFDs, traders select whether they want to take a long or short position. Traders who purchase a 'long' CFD profit if the price of the corresponding cryptoasset increases. Traders who purchase a 'short' CFD profit if the price of the corresponding investment decreases. All purchases and sales of CFDs are subject to a bid-ask spread paid to the broker who acts as the counterparty on every CFD trade.

All CFD brokers offer some form of leverage up to 20x on Bitcoin vs. the US dollar. To trade a CFD, a trader needs to post initial and maintenance margin to fund the account. The lower the margin, the higher the leverage (i.e. 10% margin requirement leads to 10x leverage). Initial Margin (IM) is the amount to fund before initiating a long or short position. The Maintenance Margin (MM) is the minimum amount of margin that traders must have against the portfolio. If the account breaches margin requirement, the broker can close out the open positions and liquidate the account. CFD traders also need to manage the daily carry cost of maintaining the positions as the daily funding rate charged by the broker ranges from 10bps to 100bps per day on the full notional amount.

In summary, cryptoassets carry significant price volatility, which can be magnified through leverage in CFDs. CFD traders also need to manage the cost of trading such as the spread, funding charges and commissions charged by the brokers. Lastly, the cryptoasset marketplace is fragmented. There can be more significant variations in the pricing of cryptoassets used to determine the value of your CFD position.

Options: An OTC derivatives market for cryptocurrency has been developing over the last few years. Since October 2017, LedgerX, a SEF and Derivatives Clearing Organization (DCO), has cleared \$130 million notional. Currently, open interest is 2,000 contracts with the longest-dated active options struck at \$15,000 and \$25,000 for December 2019 expiry.

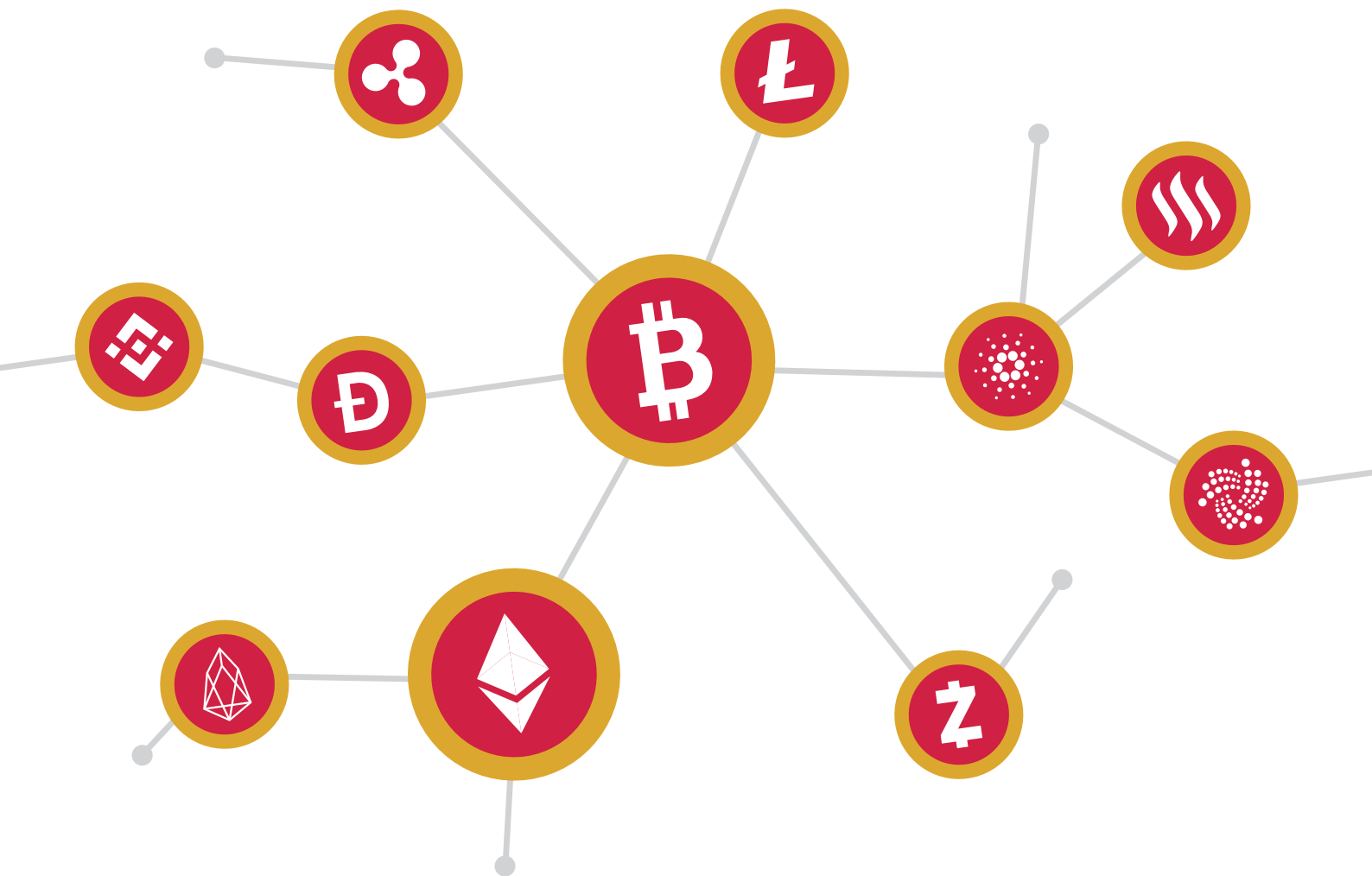
The LedgerX options product is similar to FX options. It allows a purchaser the right, but not the obligation, to purchase (in the case of a call option) or sell (in the case of a put option) Bitcoin at the price specified ('strike price') in the options contract at the expiration date. Traders who choose to write call options or put options will have the obligation to sell (in the case of a call option) or purchase (in the case of a put option) Bitcoin at the price specified in the contract at the expiration date.

However, LedgerX bitcoin options contracts are fully collateralised USD-priced European options, which are capital inefficient. Before accepting a contract for clearing, it requires traders to provide collateral to cover the maximum potential loss of the contract as well as pass a pre-trade credit check to ensure that the participant's collateral is sufficient. For a call options seller, the trader must deliver the full quantity of deliverable Bitcoin underlying the call option to LedgerX prior to entering the order. If the call is exercised, the trader's Bitcoin collateral is used to satisfy delivery. For a put options seller, the trader must deliver the full USD strike price underlying the put option to LedgerX prior to entering the order. If the put is exercised, the trader's USD collateral is used to satisfy delivery.

Summary

As more and more institutions operating across the capital markets grapple with the implications of cryptoassets, the fees and regulatory uncertainty remain a concern for traders as cryptoasset marketplaces continue to broaden the range of fiat currencies to support.

The evolution of a cryptoasset ecosystem has been a source of opportunity and obstacles and, looking back over recent years, there are lessons to be learnt. It is widely considered that products that straddle the worlds of cryptoassets and regulated derivatives exchanges are forthcoming. Indeed, the capital market for cryptoassets will continue to grow as market participants start to build institutional quality infrastructure to meet investor needs for a healthy and safe ecosystem.



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